

Total No. of Questions :6]

SEAT No. :

P201

[Total No. of Pages :2

Oct./ BE/ Insem. - 517

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS

(2015 Pattern) (Semester - I) (402044A) (Elective - I)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Figures to the right indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) Discuss the advantages and disadvantages of FEM over classical method. [4]
- b) Explain 'principle of minimum potential energy approach' to formulate FEM equations. [6]

OR

- Q2)** a) Describe the general steps of the FEM. Element Method. [4]
- b) Explain the terms. [6]
- i) Linear Static Analysis.
 - ii) Non - Linear Analysis.
- Q3)** a) Analyse the truss as shown in fig. For axial forces using finite element method. Find the displacement at node 2, Where area of cross section of element 1 is 50 mm^2 and element 2 is 40 mm^2 . [6]

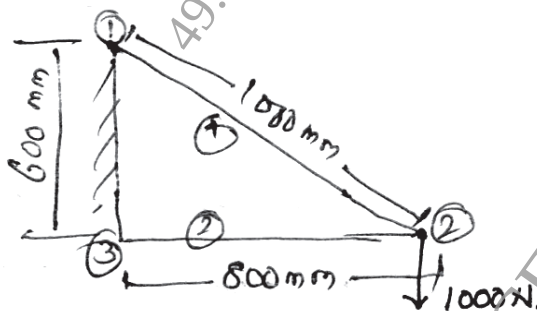


Fig.1

- b) Derive shape function in natural coordinate system of Bar. [4]

OR

P.T.O.

Q4) a) Consider the bar element as shown in fig 2, with cross sectional area $A=7.75 \times 10^{-4} \text{ M}^2$ and youngs modulus $E=2.07 \times 10^{11} \text{ N/M}^2$. If deflection at node 1 is 0.500 mm and node 2 is 0.635 mm. determine. [6]

- Displacement at point P.
- Strain E and stress σ .
- Element stiffness matrix.

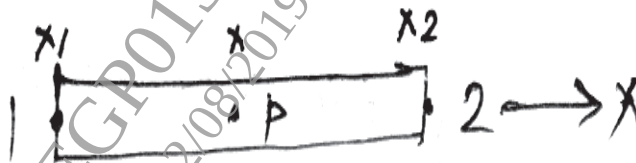


Fig.2

$$x_1 = 380 \text{ mm}$$

$$x = 508 \text{ mm}$$

$$x_2 = 585 \text{ mm}$$

b) Write short note on beam element. [4]

Q5) a) The triangular element has nodal coordinates (10,10), (40,20) and (30,50) for nodes 1,2 & 3 Resp. for a point P inside the determine the X and Y coordinates if shape function N_1 and N_2 are 0.15 and 0.25 respectively. [6]

b) Explain plane stress with example. [4]

OR

Q6) a) Write short note on polynomial shape functions. [4]

b) The nodal coordinate of triangular element are as shown in fig 3. The X coordinate, of interior point is 3.3 and shape function $N_1=0.3$ determine N_2 , N_3 and Y coord of point P. [6]

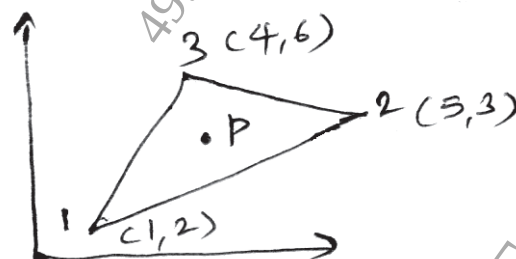


Fig.3

